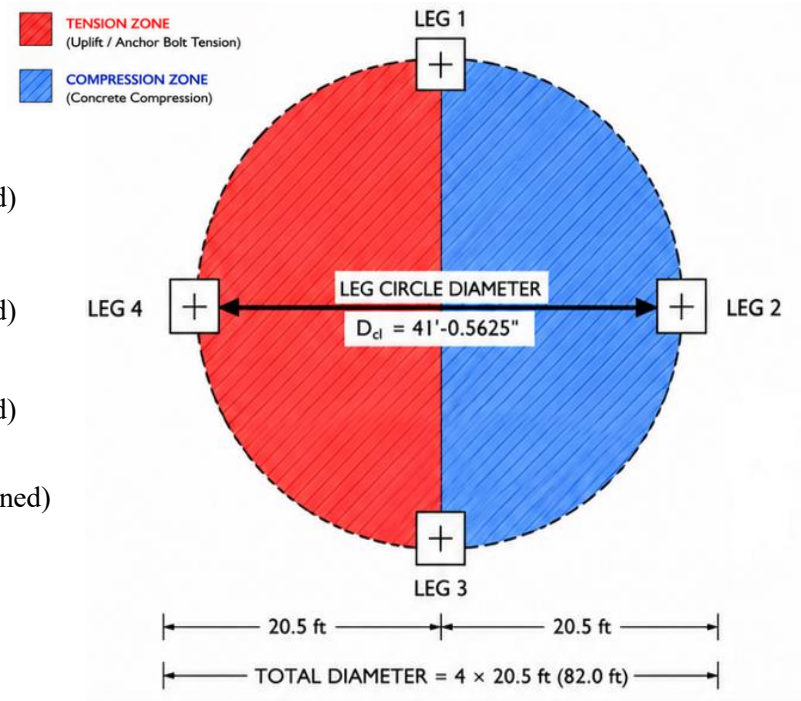
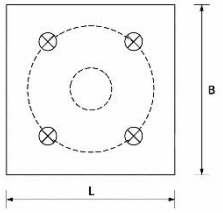
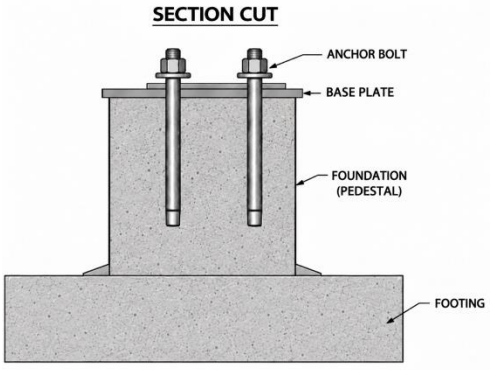
3: RINGWALL Foundation Design:

Design Parameter	Symbol	Value	Source
Number of legs	N_{leg}	**4**	User Input
Total factored axial load	P_u	**1,637.5732 kips**	Main Tank Design (Predefined)
Governing factored overturning moment	M_u	**5673.437kip-ft**	Main Tank Design (Predefined)
Total factored horizontal shear	V_u	**57.218 kips**	Main Tank Design (Predefined)
Leg circle diameter	D_{cl}	**41'-0.5625"**	Main Tank Geometry (Predefined)
Pedestal size	$B \times L$	**39 in × 39 in**	User Input
Base plate size	D_{bp}	**30 in**	User Input
Concrete compressive strength	f_c'	**4.0 ksi**	User Input / ACI 318-19
Reinforcing steel yield strength	f_y	**36 ksi**	User Input / ACI 318-19
Anchor bolts	$N_b \times d_b$	**4 × 1.5 in**	Designed in Anchor Bolts segment of this software
Anchor embedment depth	h_{ef}	**35 in**	Designed in Anchor Bolts segment of this software
Pedestal clear height used	H_p	**≈ 7'-5" = 89 in**	User Input
Concrete unit weight	γ_c	**150 pcf**	User Input







Step 1: ACI 318-19 slenderness check

ACI 318-19 6.2.5.1 permits slenderness effects to be neglected for a column not braced against sidesway when:

For a square section:

$$r = \sqrt{\frac{I}{A}}$$

I = moment of inertia of the section

r = Radius of gyration

A = Sectional Area

For a 39×39in square:

For the 39×39in square:

$$I_x = I_y = \frac{bh^3}{12}$$
$$I = \frac{39(39)^3}{12}$$
$$I = 192786.75 \text{ in}^4$$

And:

$$A = 39(39)$$
$$A = 1{,}521 \text{ in}^2$$

Therefore:

$$r = \sqrt{\frac{192786.75}{1521}}$$

r = 11.26 inch

Check A — Pedestal classification

$$\frac{\ell_u}{b} \leq 3$$
$$2.28 < 3$$

Therefore, it qualifies geometrically as a pedestal.

Check B — Structural Slenderness

$$\frac{k\ell_u}{r} \leq 22$$

Using the conservative $k = 2.0$:

$$15.81 < 22$$

Therefore, second-order slenderness effects can be neglected.

Step 2: Gravity axial load per leg

For four equally spaced legs:

$$P_{gravity} = \frac{P_u}{N_{leg}}$$
$$P_{gravity} = \frac{1637.6}{4}$$
$$\boldsymbol{P_{gravity} = 409.393 \text{ kip/leg}}$$

The pedestal self-weight can also be included.

For approximately 7'-5" pedestal height:

γ_c=150 pcf

$$W_p = (3.25)(3.25)(7.417)(0.150)$$
$$W_p \approx 11.75 \text{ kip}$$

Factored:

$$1.2W_p = 14.10 \text{ kip}$$

Therefore, conservatively:

$$P_{comp,total} = 409.393 + 14.10$$
$$\boldsymbol{P_{comp,total} \approx 423.494 \text{ kip}}$$

Step 4: Pedestal Concrete Bearing Check

Base plate:

$$A_1 = \frac{\pi(30)^2}{4} = 706.86 \text{ in}^2$$

Pedestal:

$$A_2 = 39(39) = 1521 \text{ in}^2$$

Bearing enhancement factor:

$$\sqrt{\frac{A_2}{A_1}} = \sqrt{\frac{1521}{706.86}} = 1.467$$

Using the concrete bearing-strength relationship: $\phi B_n = \phi(0.85 f_c' A_1)\sqrt{\frac{A_2}{A_1}}$

Taking:

Transverse Pedestal Reinforcement: if

1. Spiral = 0.75
2. Ties = 0.65

$$\phi P_n = 0.65(0.85)(4)(706.86)(1.30)$$

$$\boldsymbol{\phi P_n \approx 2291.517 \text{ kips}}$$

Demand:

$$P_u = \mathbf{423.494} \text{ kip}$$

$$\mathbf{D/C = \frac{423.494}{2292}}$$

Therefore:

$0.185 < 1.00$……. **CONCRETE BEARING — PASS**

Demand/capacity ratio:……. $D/C \approx 0.20$

So the **39 × 39 in pedestal is more than adequate for bearing**.

Step 5: Pedestal longitudinal Reinforcement

$$\mathbf{18 - \#6 \text{ vertical bars}}$$

Area of one #6:

$$A_b = 0.44 \text{ in}^2$$

Therefore:

$$A_s = 18(0.44)$$

$$\boldsymbol{A_s = 7.92 \text{ in}^2}$$

$$A_g = 39(39)$$

$$A_s = 1521\ in^2$$

Reinforcement ratio:

$$\rho = \frac{7.92}{39(39)}$$

$$\rho = 0.0053$$

or:……$\boldsymbol{\rho = 0.53\%}$

For the cast-in-place pedestal-to-foundation interface, ACI 318-19 requires reinforcement crossing the interface of at least:

$$A_{s,min} = 0.005A_g$$

ACI 318-19 Section 16.3.4.1 specifies the $0.005A_g$ minimum for connections between a cast-in-place column or pedestal and its foundation.

As,min=0.001(1521)

$$A_{s,min} = 7.605\ in^2$$

Provided:

$$A_s = 7.92\ \text{in}^2$$

Therefore:

$$7.92 > 7.605\ldots.\ \textbf{PASS}$$

Reinforcement ratio:

$$\rho_g = \frac{7.92}{1{,}521}$$
$$\rho_g = 0.00521 = 0.521\%$$

5:

$$A_g = 39(39) = 1521\ \text{in}^2$$
$$A_{s,\text{min}} = 0.005A_g$$
$$A_{s,\text{min}} = 0.005(1521) = 7.605\ \text{in}^2$$

Provided:

$$A_s = 18(0.44) = 7.92\ \text{in}^2$$

$$\frac{A_s}{A_{s,\text{min}}} = \frac{7.92}{7.605} = 1.041$$

$$A_s > A_{s,\min} \ldots\ldots\ldots \text{ OK}$$

Step 6: Axial Compression Capacity

Gross area:

$$A_g = 1{,}521 \text{ in}^2$$

Steel area:

$$A_s = 7.92 \text{ in}^2$$

Nominal concentric compression strength:

ACI 318-14 21.2.2, Table 21.2.2 — Strength reduction factor ϕ for moment, axial force, or combined moment and axial force.

Transverse reinforcement ϕ (Mehul; we need to add this table Infront of ϕ that user can select the shape and based on it the factor can be populated).

No	ϕ
1. Spirals	0.75
2. Tied	0.65

$P_o = 0.85f_c'(A_g - A_s) + f_y A_s$
$P_o = 0.85(4.0)(1,521 - 7.92) + 60(7.92)$
$\boldsymbol{P_o = 5,429.592 \text{ kips}}$

Using a conservative tied-member design strength factor and 0.80 maximum axial-strength factor:

$$\phi P_n = 0.65(0.80)(5,429.592)$$
$$\boldsymbol{\phi P_n = 2,823.388 \text{ kips}}$$

Then for **spiral reinforcement**:

$$P_n = 0.85(5,429.592)$$
$$P_n = 4{,}615.153 \ kips$$

Finally:

$$\phi P_n = 0.75(4,615.153)$$
$$\phi P_n = 3461.365\ kips$$

Demand is:

Therefore:

$$\phi P_n = 3{,}461.365\ kips > P_u = 423.494\ kips$$

and the utilization is:

$$\frac{P_u}{\phi P_n} = \frac{423.494}{3461.365} = 0.122$$

Demand:

$$P_u = \mathbf{478.971}\ \text{kips}$$

Therefore:

$$\text{D/C} = \frac{\mathbf{478.971}}{2{,}922.2}$$
$$\mathbf{D/C = 0.17}$$

Step 7: Pedestal Tensile Capacity Check

Governing tension-side pedestal demand:

$$\boldsymbol{T_u = 207.329\ \text{kips}}$$

longitudinal reinforcement:

$$A_s = 7.92\ \text{in}^2$$

Using:

$$\phi = 0.90$$

and:

$$f_y = 60 \text{ ksi}$$

The available steel tension strength is:

$$\phi T_n = 0.90(7.92)(36)$$
$$\boldsymbol{\phi T_n = 256.608 \text{ kips}}$$

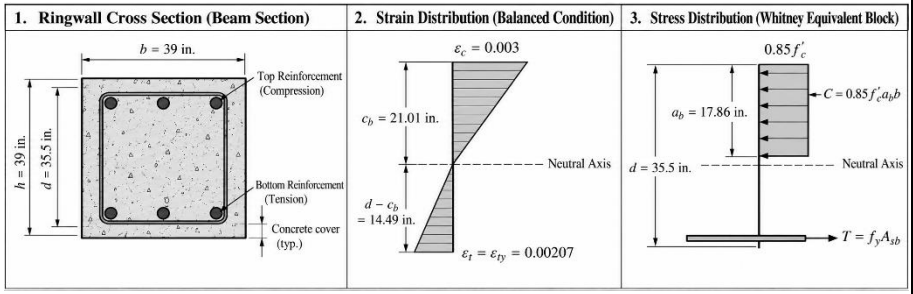
Demand/capacity ratio:

$$\text{D/C} = \frac{207.329}{256.608}$$
$$\textbf{D/C} = \mathbf{0.808}\textbf{.....}\text{PASS}$$

Therefore, the 18-#6 vertical reinforcement has adequate theo retical tensile capacity for the pedestal uplift-transfer demand.

Step 8: Pedestal *P-M* Interaction Check

8.1 Governing Axial Load and Moment



$$\boldsymbol{b = 39 \text{ in.}}$$
$$\boldsymbol{d = 35.5 \text{ in.}}$$
$$\boldsymbol{f'_c = 4 \text{ ksi}}$$
$$\boldsymbol{f_y = 60 \text{ ksi}}$$

Assume:

$$E_s = 29{,}000 \text{ ksi}$$

1. Steel yield strain

$$\epsilon_{ty} = \frac{f_y}{E_s} = \frac{60}{29{,}000}$$
$$\epsilon_{ty} = 0.00207$$

2. Find balanced neutral-axis depth c_b

Using the same similar-triangle equation from your professor:

$$c_b = d\frac{0.003}{0.003 + \epsilon_{ty}}$$
$$c_b = 35.5\frac{0.003}{0.003 + 0.00207}$$
$$c_b = 21.01 \text{ in.}$$

This is calculated from your section—it is not assumed.

3. Find a_b

For:

$$f_c' = 4 \text{ ksi}$$
$$\beta_1 = 0.85$$

Therefore:

$$a_b = \beta_1 c_b$$
$$a_b = 0.85(21.01)$$
$$a_b = 17.86 \text{ in.}$$

4. Balanced reinforcement A_{sb}

From equilibrium:

$$C = T$$
$$0.85 f_c' a_b b = f_y A_{sb}$$

Therefore:

$$A_{sb} = \frac{0.85 f_c' a_b b}{f_y}$$
$$A_{sb} = \frac{0.85(4)(17.86)(39)}{60}$$

$$A_{sb} = 39.47 \text{ in}^2$$

5. Balanced tensile force

$$T = f_y A_{sb}$$
$$T = 60(39.47)$$
$$T = 2{,}368.1 \text{ kips}$$

6. Balanced nominal moment M_{nb}

Same equation as your professor's example:

$$M_{nb} = T\left(d - \frac{a_b}{2}\right)$$

Convert in-kips to kip-ft:

$$M_{nb} = \frac{2368.1\left(35.5 - \frac{17.86}{2}\right)}{12}$$
$$M_{nb} = 5{,}243.4 \text{ kip-ft}$$

Final answer

$$c_b = 21.01 \text{ in.}$$
$$a_b = 17.86 \text{ in.}$$
$$A_{sb} = 39.47 \text{ in}^2$$
$$M_{nb} = 5{,}243.4 \text{ kip-ft}$$

$\phi = 0.65$:

$$\phi M_{nb} = 0.65(5243.4)$$
$$\boldsymbol{\phi M_{nb} = 3{,}408.2 \text{ kip-ft}}$$

Step 9: Horizontal Shear per Pedestal

Total tank shear:

$$V_u = 57.218 \text{ kip}$$

Assuming equal distribution among four legs:

$$V_{u,ped} = \frac{57.218}{4}$$
$$\boldsymbol{V_{u,ped} = 14.305 \text{kip}}$$

This is consistent with the anchor design, which subsequently distributes the pedestal shear among four anchor bolts and obtains 3.27 kip/bolt.

Step 9: Pedestal Tie Design

1. Tie diameter

25.7.2.2 Diameter of tie bar shall be at least (a) or (b):

(a) No. 3 enclosing No. 10 or smaller longitudinal bars
(b) No. 4 enclosing No. 11 or larger longitudinal bars or bundled longitudinal bars

ACI 318-19 25.7.2.2: For longitudinal bars #10 or smaller:

$$d_{bt} \geq \#3$$

Therefore:

$$d_{bt} = \#3$$

ACI 318-19 25.7.2.1 requires:

$$s \leq \min(16, d_b,\ 48d_{bt}\ b_{\min})$$

Therefore:

$$16d_b = 16(1.00) = 16 \text{ in}$$
$$48d_{bt} = 48(0.375) = 18 \text{ in}$$
$$b_{\min} = 39 \text{ in}$$

Thus:

$$s_{\max} = \min(16, 18, 39)$$
$$s_{\max} = 16 \text{ in}$$

Adopt:

#3 closed ties @ 12 in. o.c.

This provides: **12<16 in**

Final detail

Pedestal: 20-#8 vertical + #3 ties/crossties @ 12" o.c.

with the top 24 in. optionally detailed at 6 in. o.c. around the anchor/base-plate zone.

Step 10: Pedestal-to-Footing Development

ACI 318-19 25.4.2.

Tension development-length expression:

$$l_d = \frac{3}{40}\frac{f_y}{\lambda\sqrt{f_c'}}\frac{\psi_t\psi_e\psi_s\psi_g}{(c_b + K_{tr})/d_b}d_b$$

$\lambda = 1.0$
$f_y = 60{,}000$ psi
$f_c' = 4{,}000$ psi
$\psi_t = \psi_e = \psi_s = \psi_g = 1.0$

and conservatively:

$$\frac{c_b + K_{tr}}{d_b} = 2.5$$

Then:

$$l_d = \frac{3}{40}\frac{60{,}000}{\sqrt{4{,}000}}\frac{1.0}{2.5}(1.00)$$
$l_d \approx 28.46$ in

The ACI minimum development length is:

$$l_d \geq 12 \text{ in}$$

Therefore:

$$l_d \approx 28.5 \text{ in}$$

Adopt:

$$l_d = 30 \text{ in minimum}$$

11.2 Development at the pedestal-footing interface

The pedestal longitudinal bars shall extend into the footing by at least:

$$30 \text{ in}$$

or the greater length required by the final ACI 318-19 development calculation based on the actual bar location and footing geometry.

The provided development length is therefore:

$$l_{prov} = 30 \text{ in}$$

and:

$$\frac{l_{prov}}{l_d} = \frac{30}{28.5} = 1.05$$

Thus:

$$l_{prov} \geq l_d \ \therefore \text{OK}$$

Factor	Condition	Value
ψ_t**Casting position**	More than 12 in. fresh concrete below horizontal bar	**1.3**
	Other conditions	**1.0**
ψ_e**Epoxy/coating**	Epoxy-coated, cover < $3d_b$ **or** spacing < $6d_b$	**1.5**

Factor	Condition	Value
	Epoxy-coated, other conditions	**1.2**
	Uncoated or galvanized	**1.0**
ψ_s **Bar size**	#6 and smaller	**0.8**
	#7 and larger	**1.0**
ψ_g **Steel grade**	Grade 40 or 60	**1.0**
	Grade 80	**1.15**
	Grade 100	**1.30**

ACI 318-19 (25.4.2.5)